1.
빌드 시스템과 빌드 툴의 차이를 설명하라.
답안:
- 빌드 시스템: 소스 코드를 실행 가능한 프로그램으로 만드는 전체 과정과 규칙을 관리하는 체계이다.
- 빌드 툴: 빌드 시스템에서 실제 빌드 작업을 수행하거나 빌드 과정을 생성·관리하는 도구이다.
예를 들어 ROS2에서는 ament_cmake가 빌드 시스템 역할을 하고, colcon이 여러 패키지의 빌드를 관리하는 도구로 사용된다.

1-2.



1-3.
ROS2에서 새로운 패키지를 생성할 때는 ros2 pkg create 명령어를 사용한다.

ros2 pkg create --build-type ament_cmake first_pkg

- ros2 pkg create: 새로운 ROS2 패키지를 생성한다.
- --build-type: 사용할 빌드 유형을 지정한다.
- ament_cmake: CMake 기반 빌드 방식을 사용한다.
- first_pkg: 생성할 패키지 이름이다.
명령어를 실행하면 first_pkg 디렉터리가 생성되고, 그 안에 package.xml, CMakeLists.txt 등의 파일이 만들어진다.



1-4.



1-5.
ROS2에서는 colcon build 명령어를 사용하여 패키지를 빌드한다.

전체 패키지 빌드:

```
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ colcon build
Starting >>> first_pkg
Finished <<< first_pkg [2.09s]

Summary: 1 package finished [2.40s]
```

특정 패키지만 빌드:

```
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ colcon build --packages-select first_pkg
Starting >>> first_pkg
Finished <<< first_pkg [0.33s]

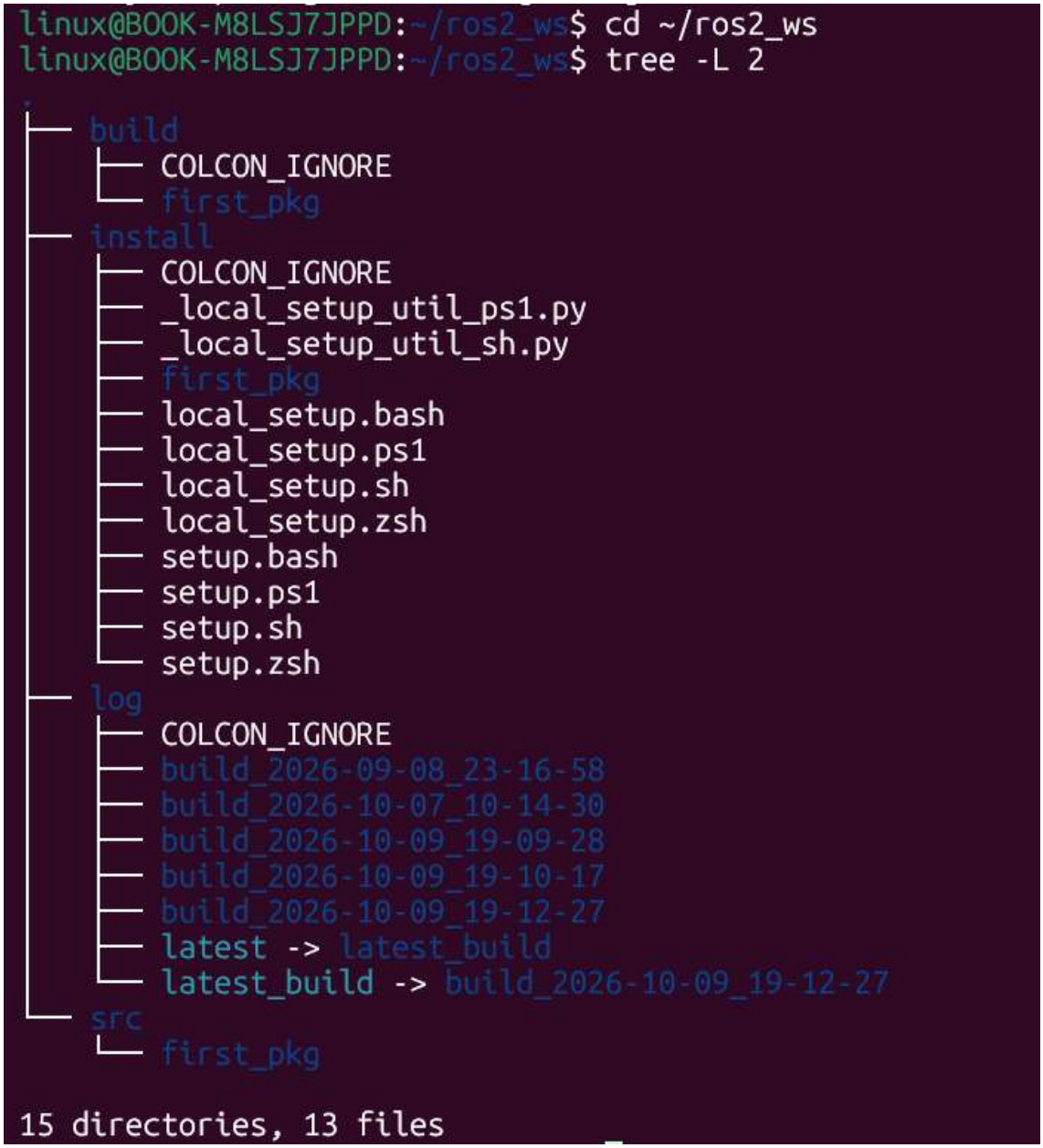
Summary: 1 package finished [0.77s]
```

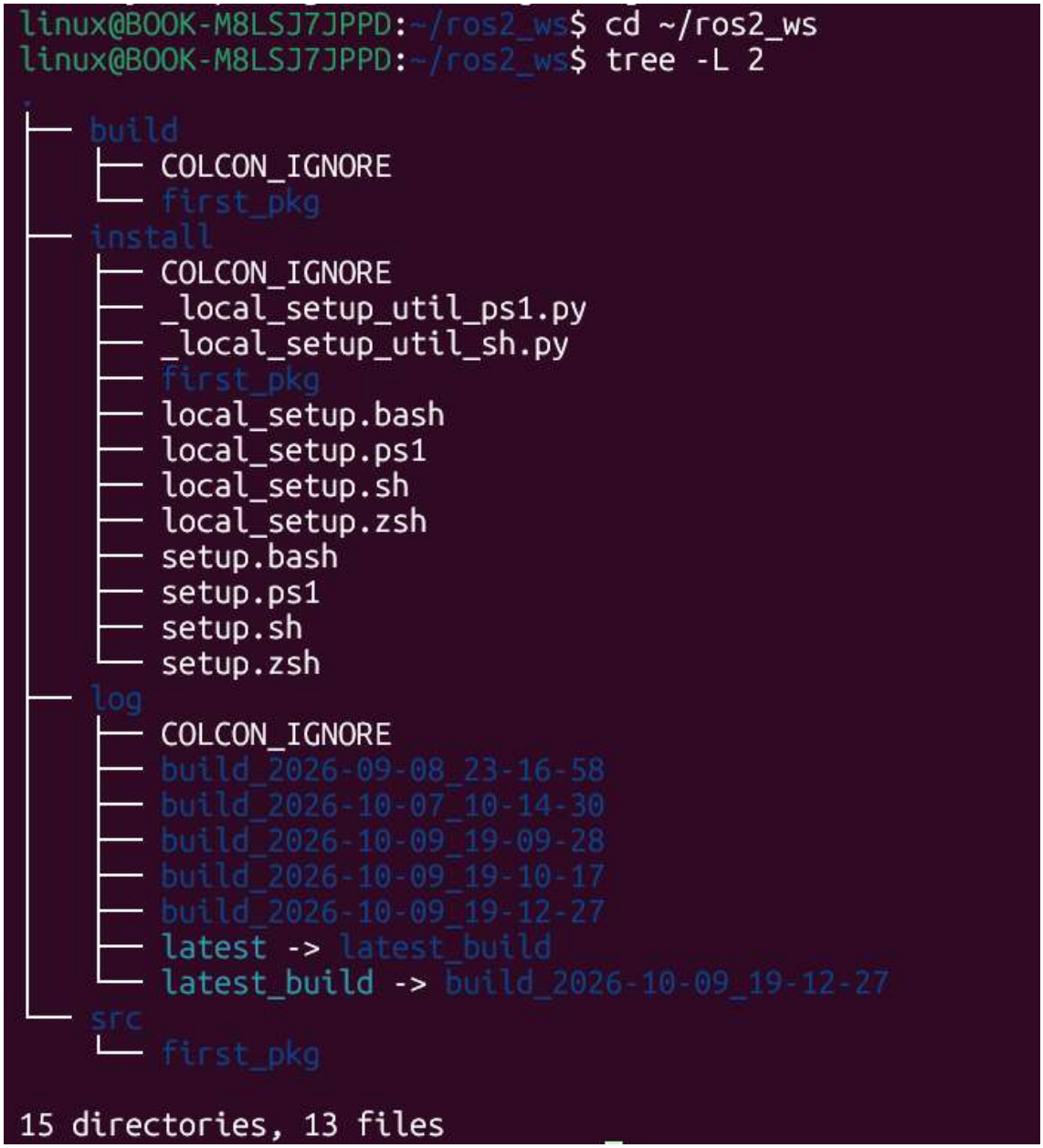
2-1

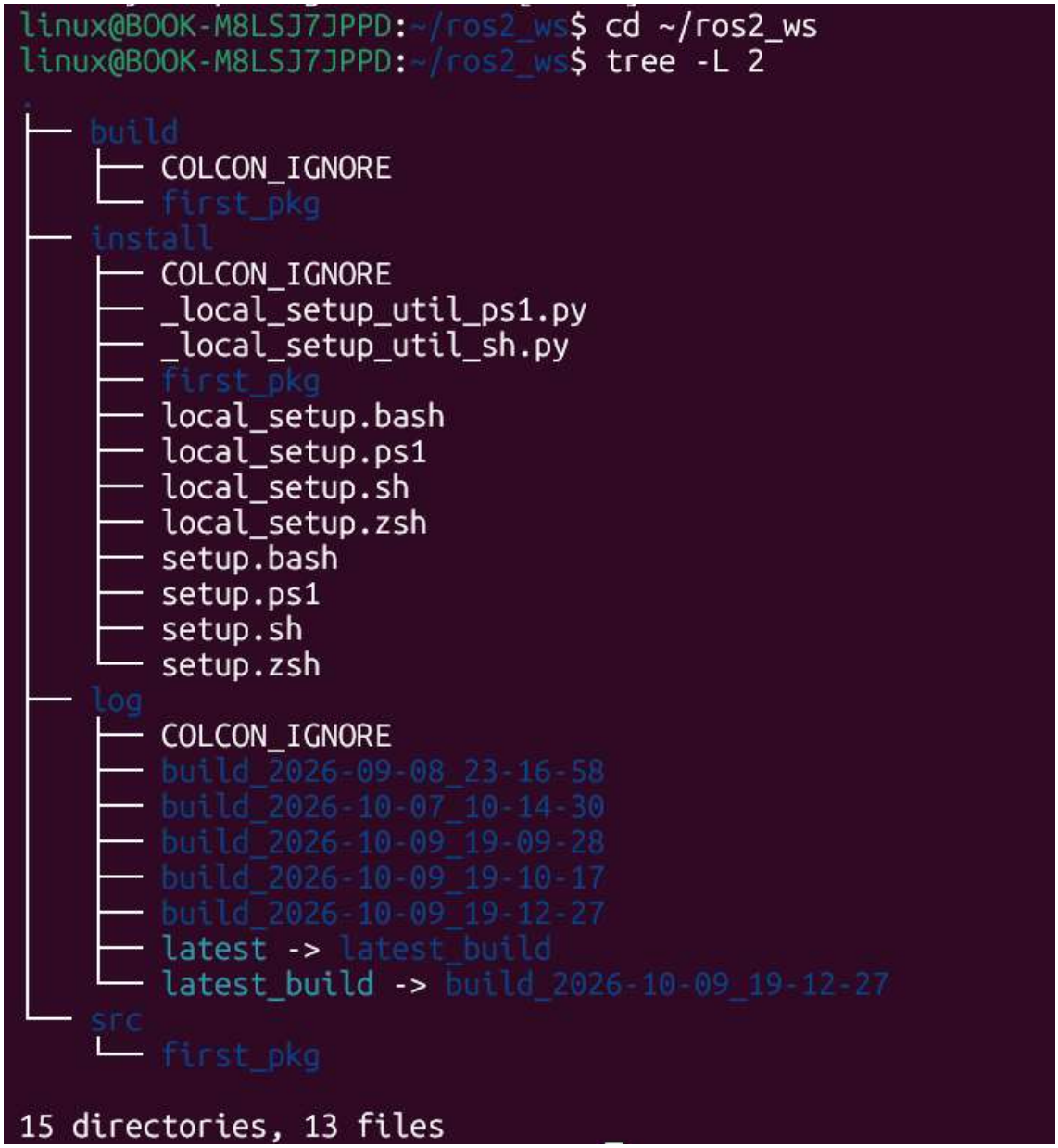
```
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ cd ~/ros2_ws
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ pwd
/home/linux/ros2_ws
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ ls
build  install  log  src
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ cd ~/ros2_ws/src
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ ls
first_pkg
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ cd ~/ros2_ws
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ colcon build --packages-select first_pkg
Starting >>> first_pkg
Finished <<< first_pkg [0.13s]

Summary: 1 package finished [0.32s]
```

2-2


linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ cd ~/ros2_ws
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ tree -L 2
.
├── build
│   ├── COLCON_IGNORE
│   └── first_pkg
├── install
│   ├── COLCON_IGNORE
│   ├── _local_setup_util_ps1.py
│   ├── _local_setup_util_sh.py
│   ├── first_pkg
│   ├── local_setup.bash
│   ├── local_setup.ps1
│   ├── local_setup.sh
│   ├── local_setup.zsh
│   ├── setup.bash
│   ├── setup.ps1
│   ├── setup.sh
│   └── setup.zsh
├── log
│   ├── COLCON_IGNORE
│   ├── build_2026-09-08_23-16-58
│   ├── build_2026-10-07_10-14-30
│   ├── build_2026-10-09_19-09-28
│   ├── build_2026-10-09_19-10-17
│   ├── build_2026-10-09_19-12-27
│   ├── latest -> latest_build
│   └── latest_build -> build_2026-10-09_19-12-27
└── src
    └── first_pkg
15 directories, 13 files


linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ cd ~/ros2_ws
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ tree -L 2
.
├── build
│   ├── COLCON_IGNORE
│   └── first_pkg
├── install
│   ├── COLCON_IGNORE
│   ├── _local_setup_util_ps1.py
│   ├── _local_setup_util_sh.py
│   ├── first_pkg
│   ├── local_setup.bash
│   ├── local_setup.ps1
│   ├── local_setup.sh
│   ├── local_setup.zsh
│   ├── setup.bash
│   ├── setup.ps1
│   ├── setup.sh
│   └── setup.zsh
├── log
│   ├── COLCON_IGNORE
│   ├── build_2026-09-08_23-16-58
│   ├── build_2026-10-07_10-14-30
│   ├── build_2026-10-09_19-09-28
│   ├── build_2026-10-09_19-10-17
│   ├── build_2026-10-09_19-12-27
│   ├── latest -> latest_build
│   └── latest_build -> build_2026-10-09_19-12-27
└── src
    └── first_pkg
15 directories, 13 files


linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ cd ~/ros2_ws
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ tree -L 2
.
├── build
│   ├── COLCON_IGNORE
│   └── first_pkg
├── install
│   ├── COLCON_IGNORE
│   ├── _local_setup_util_ps1.py
│   ├── _local_setup_util_sh.py
│   ├── first_pkg
│   ├── local_setup.bash
│   ├── local_setup.ps1
│   ├── local_setup.sh
│   ├── local_setup.zsh
│   ├── setup.bash
│   ├── setup.ps1
│   ├── setup.sh
│   └── setup.zsh
├── log
│   ├── COLCON_IGNORE
│   ├── build_2026-09-08_23-16-58
│   ├── build_2026-10-07_10-14-30
│   ├── build_2026-10-09_19-09-28
│   ├── build_2026-10-09_19-10-17
│   ├── build_2026-10-09_19-12-27
│   ├── latest -> latest_build
│   └── latest_build -> build_2026-10-09_19-12-27
└── src
    └── first_pkg
15 directories, 13 files

```
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ cd ~/ros2_ws
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ tree -L 2
.
├── build
│   ├── COLCON_IGNORE
│   └── first_pkg
├── install
│   ├── COLCON_IGNORE
│   ├── _local_setup_util_ps1.py
│   ├── _local_setup_util_sh.py
│   ├── first_pkg
│   ├── local_setup.bash
│   ├── local_setup.ps1
│   ├── local_setup.sh
│   ├── local_setup.zsh
│   ├── setup.bash
│   ├── setup.ps1
│   ├── setup.sh
│   └── setup.zsh
├── log
│   ├── COLCON_IGNORE
│   ├── build_2026-09-08_23-16-58
│   ├── build_2026-10-07_10-14-30
│   ├── build_2026-10-09_19-09-28
│   ├── build_2026-10-09_19-10-17
│   ├── build_2026-10-09_19-12-27
│   ├── latest -> latest_build
│   └── latest_build -> build_2026-10-09_19-12-27
└── src
    └── first_pkg

15 directories, 13 files
```

```
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ tree -L 2 ~/ros2_ws/src/first_pkg
/home/linux/ros2_ws/src/first_pkg
├── CMakeLists.txt
├── include
│   └── first_pkg
├── package.xml
└── src

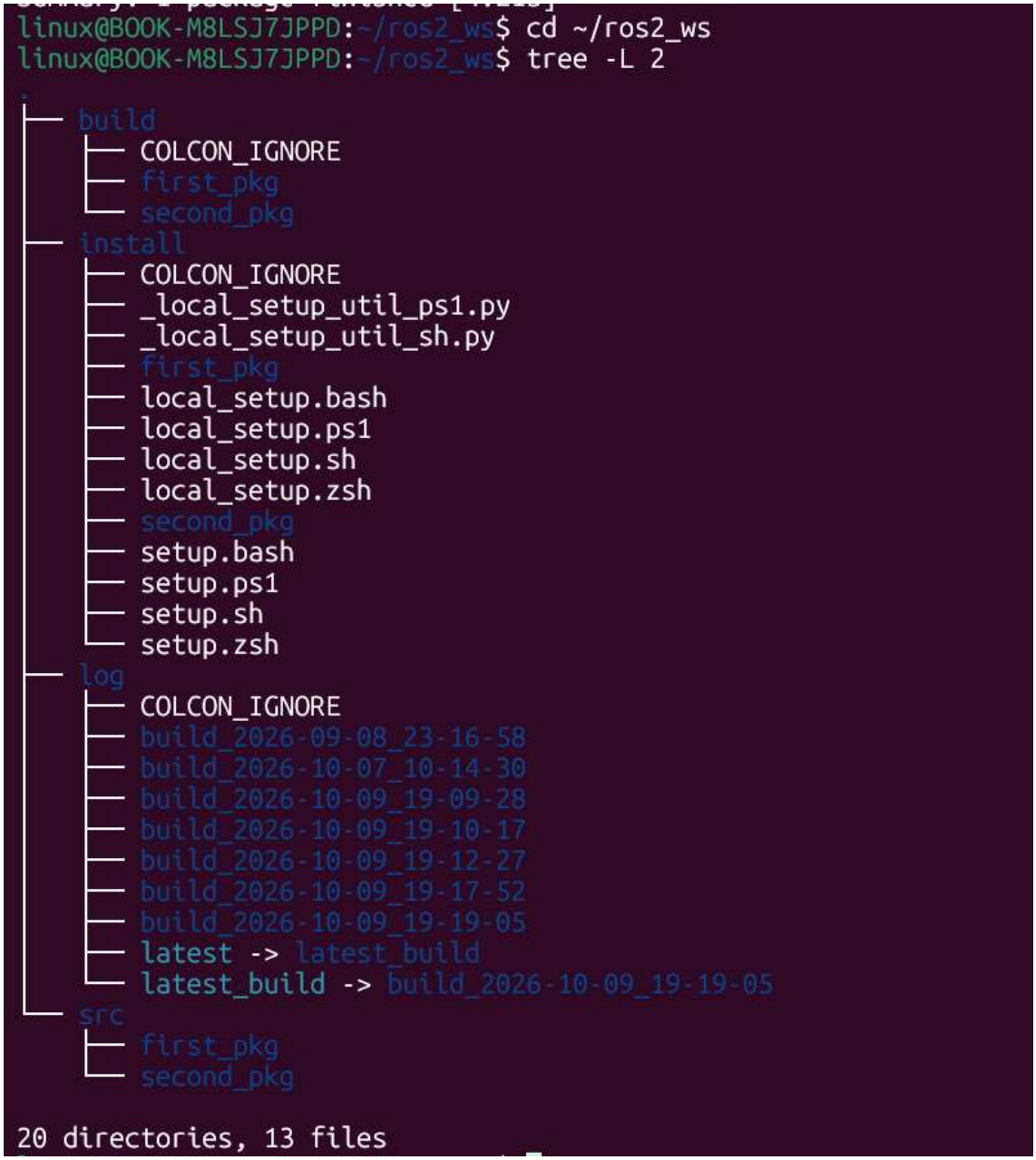
4 directories, 2 files
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ colcon build --packages-select first_pkg
Starting >>> first_pkg
Finished <<< first_pkg [0.18s]

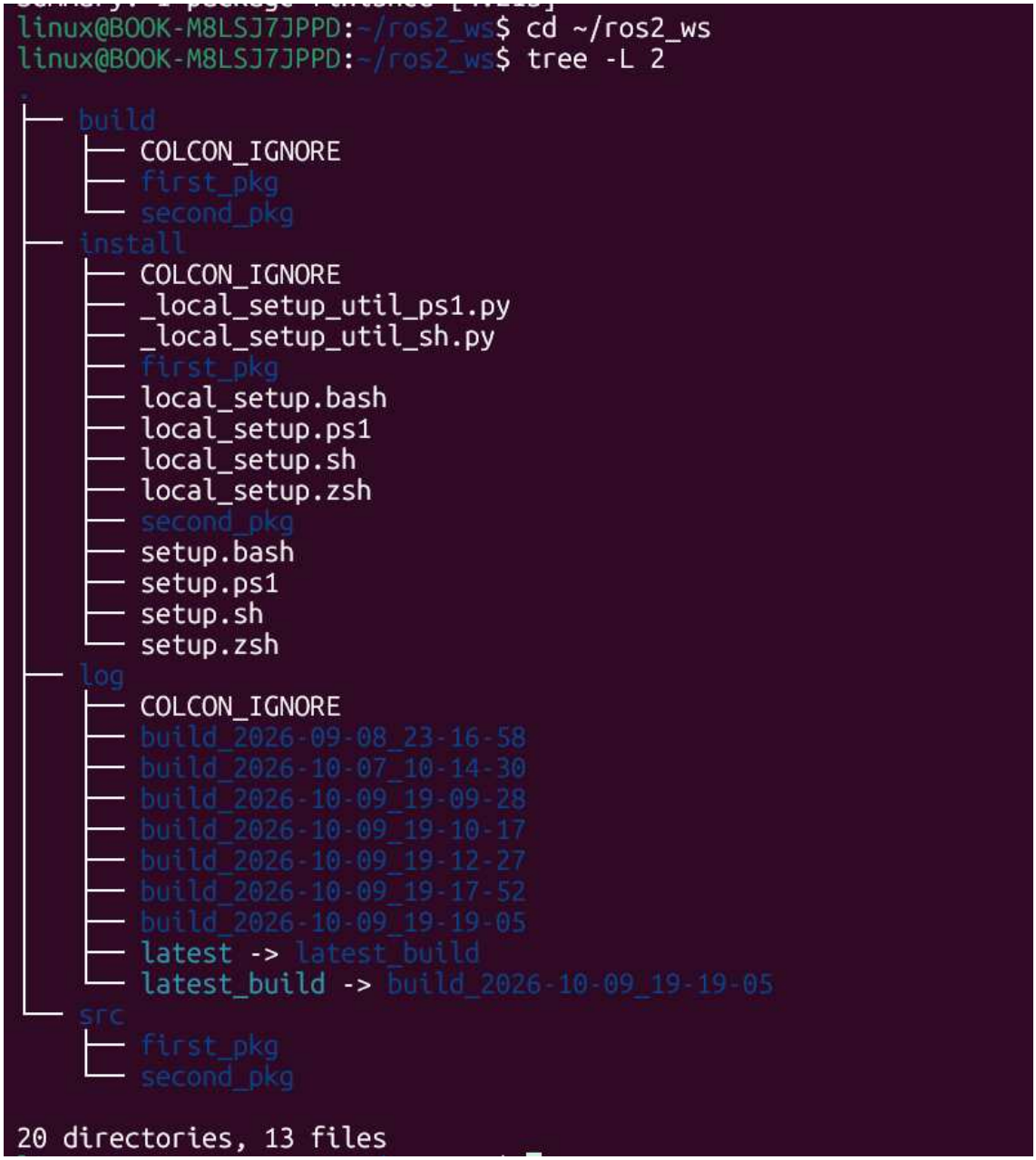
Summary: 1 package finished [0.50s]
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$
```

```
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ cd ~/ros2_ws/src
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ ros2 pkg create --build-type ament_cmake second_pkg
going to create a new package
package name: second_pkg
destination directory: /home/linux/ros2_ws/src
package format: 3
version: 0.0.0
description: TODO: Package description
maintainer: ['linux <linux@todo.todo>']
licenses: ['TODO: License declaration']
build type: ament_cmake
dependencies: []
creating folder ./second_pkg
creating ./second_pkg/package.xml
creating source and include folder
creating folder ./second_pkg/src
creating folder ./second_pkg/include/second_pkg
creating ./second_pkg/CMakeLists.txt

[WARNING]: Unknown license 'TODO: License declaration'.  This has been set in the package.xml, but no LICENSE file has been created.
It is recommended to use one of the ament license identifiers:
Apache-2.0
BSL-1.0
BSD-2.0
BSD-2-Clause
BSD-3-Clause
GPL-3.0-only
LGPL-3.0-only
MIT
MIT-0
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ cd ~/ros2_ws
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ colcon build --packages-select second_pkg
Starting >>> second_pkg
Finished <<< second_pkg [4.01s]

Summary: 1 package finished [4.21s]
```


linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ cd ~/ros2_ws
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ tree -L 2
.
├── build
│   ├── COLCON_IGNORE
│   ├── first_pkg
│   └── second_pkg
├── install
│   ├── COLCON_IGNORE
│   ├── _local_setup_util_ps1.py
│   ├── _local_setup_util_sh.py
│   ├── first_pkg
│   ├── local_setup.bash
│   ├── local_setup.ps1
│   ├── local_setup.sh
│   ├── local_setup.zsh
│   ├── second_pkg
│   ├── setup.bash
│   ├── setup.ps1
│   ├── setup.sh
│   └── setup.zsh
├── log
│   ├── COLCON_IGNORE
│   ├── build_2026-09-08_23-16-58
│   ├── build_2026-10-07_10-14-30
│   ├── build_2026-10-09_19-09-28
│   ├── build_2026-10-09_19-10-17
│   ├── build_2026-10-09_19-12-27
│   ├── build_2026-10-09_19-17-52
│   ├── build_2026-10-09_19-19-05
│   ├── latest -> latest_build
│   └── latest_build -> build_2026-10-09_19-19-05
└── src
    ├── first_pkg
    └── second_pkg
20 directories, 13 files


linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ cd ~/ros2_ws
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ tree -L 2
.
├── build
│   ├── COLCON_IGNORE
│   ├── first_pkg
│   └── second_pkg
├── install
│   ├── COLCON_IGNORE
│   ├── _local_setup_util_ps1.py
│   ├── _local_setup_util_sh.py
│   ├── first_pkg
│   ├── local_setup.bash
│   ├── local_setup.ps1
│   ├── local_setup.sh
│   ├── local_setup.zsh
│   ├── second_pkg
│   ├── setup.bash
│   ├── setup.ps1
│   ├── setup.sh
│   └── setup.zsh
├── log
│   ├── COLCON_IGNORE
│   ├── build_2026-09-08_23-16-58
│   ├── build_2026-10-07_10-14-30
│   ├── build_2026-10-09_19-09-28
│   ├── build_2026-10-09_19-10-17
│   ├── build_2026-10-09_19-12-27
│   ├── build_2026-10-09_19-17-52
│   ├── build_2026-10-09_19-19-05
│   ├── latest -> latest_build
│   └── latest_build -> build_2026-10-09_19-19-05
└── src
    ├── first_pkg
    └── second_pkg
20 directories, 13 files